40th 1978-2017 Australian Mathematics Competition

AUSTRALIAN MATHEMATICS TRUST

# AUSTRALIAN MATHEMATICS COMPETITION

AN ACTIVITY OF THE AUSTRALIAN MATHEMATICS TRUST

NAME _____

YEAR _____ TEACHER _____

# 2017
# SIMPLIFIED CHINESE VERSION
# JUNIOR DIVISION

## AUSTRALIAN SCHOOL YEARS 7 and 8

TIME ALLOWED: 75 MINUTES

## INSTRUCTIONS AND INFORMATION

### GENERAL

1. Do not open the booklet until told to do so by your teacher.
2. NO calculators, maths stencils, mobile phones or other calculating aids are permitted. Scribbling paper, graph paper, ruler and compasses are permitted, but are not essential.
3. Diagrams are NOT drawn to scale. They are intended only as aids.
4. There are 25 multiple-choice questions, each requiring a single answer, and 5 questions that require a whole number answer between 0 and 999. The questions generally get harder as you work through the paper. There is no penalty for an incorrect response.
5. This is a competition not a test; do not expect to answer all questions. You are only competing against your own year in your own country/Australian state so different years doing the same paper are not compared.
6. Read the instructions on the answer sheet carefully. Ensure your name, school name and school year are entered. It is your responsibility to correctly code your answer sheet.
7. When your teacher gives the signal, begin working on the problems.

### THE ANSWER SHEET

1. Use only lead pencil.
2. Record your answers on the reverse of the answer sheet (not on the question paper) by FULLY colouring the circle matching your answer.
3. Your answer sheet will be scanned. The optical scanner will attempt to read all markings even if they are in the wrong places, so please be careful not to doodle or write anything extra on the answer sheet. If you want to change an answer or remove any marks, use a plastic eraser and be sure to remove all marks and smudges.

### INTEGRITY OF THE COMPETITION

The AMT reserves the right to re-examine students before deciding whether to grant official status to their score.

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## 1-10 题，每题 3 分
## Questions 1 to 10, 3 marks each

**1.** 算式 $2+0+1+7$ 之值等于

The value of $2+0+1+7$ is

(A) 10 (B) 19 (C) 37 (D) 208 (E) 2017

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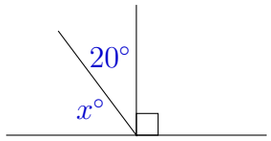
**2.** 在右图中，请问 $x$ 之值是什么？

What is the value of $x$ in the diagram?




(A) 20 (B) 70 (C) 80 (D) 110 (E) 160

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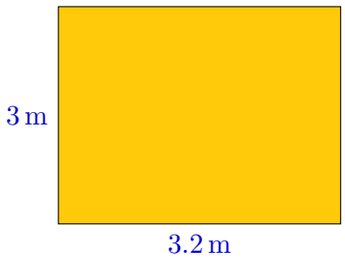
**3.** 在右图中，矩形的长为 3.2 m、宽为 3 m。则这个矩形的面积等于

This rectangle is 3.2 m wide and 3 m tall. Its area is




(A) $9.6\,\mathrm{m}^2$ (B) $15\,\mathrm{m}^2$ (C) $90.6\,\mathrm{m}^2$ (D) $9.2\,\mathrm{m}^2$ (E) $6.5\,\mathrm{m}^2$

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**4.** 从 13 开始数起，接着每个数都增加 5，所数出的数为 13、18、23 $\cdots$。请问下列哪一个选项之中的数是您将会数出的数之一？

Starting with 13, and counting by fives, you count $13, 18, 23$, and so on. Which one of the following numbers will be one of the numbers you count?

(A) 47 (B) 48 (C) 49 (D) 50 (E) 51

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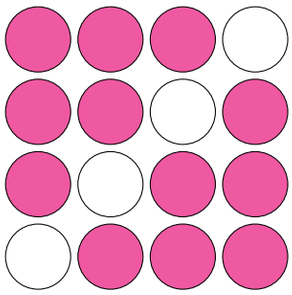
**5.** 请问图中有几分之几的圆圈被涂上阴影？

What fraction of these circles are shaded?



(A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{2}{3}$ (D) $\frac{3}{4}$ (E) $\frac{9}{16}$

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**6.** 请问下列哪一个选项中的数码填入下面的方框内使得所构成的这个三位数可被 3 整除？

Of the following, which digit could be put in the box to make this three-digit number divisible by 3?

$$1\,\square\,7$$

(A) 1 (B) 3 (C) 6 (D) 8 (E) 9

---

**7.** 一个帮浦每运行 150 分钟需用掉 8 L 的生质柴油。请问这个帮浦使用 32 L 的生质柴油可以运行几个小时？

A pump runs for 150 minutes, using 8 litres of biodiesel. For how many hours can it run with 32 litres of biodiesel?

(A) 6 (B) 7 (C) 8 (D) 10 (E) 12

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**8.** 小强从超市帮妈妈提一个购物袋回家，袋子内有 780 g 的鱼、1.35 kg 的蔬菜与 680 g 的水果。已知这个购物袋本身的重量为 150 g，请问小强所提的东西总重为多少 kg？

Jonah returned from the shop with a bag carrying 780 g of fish, 1.35 kg of vegetables, and 680 g of fruit for his mother. The bag itself weighed 150 g. The total weight, in kilograms, that Jonah carried was

(A) 1.745 (B) 2 (C) 2.81 (D) 2.96 (E) 3

---

**9.** 1 的 1000% 等于

1000% of 1 is

(A) 0.1 (B) 1 (C) 10 (D) 100 (E) 1000

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**10.** 请问下列哪一个选项中的数填入下面的方框内可使得所构成的分数之值介于 7 与 8 之间？

Which one of the following numbers could be put in the box to make the fraction between 7 and 8?

$$\frac{\square + 3}{6}$$

(A) 47 (B) 25 (C) 32 (D) 37 (E) 41

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## 11-20 题，每题 4 分
## Questions 11 to 20, 4 marks each

**11.** 小艾玩文字游戏。每拨动老爷钟一下，她就将两个字母互换位置。请问她最少要拨动几下老爷钟才能将 WORDS 变成 SWORD？

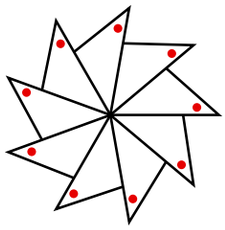
Alice is playing with words. At each tick of her grandfather's clock she swaps two letters. What is the smallest number of clock ticks during which she can change WORDS to SWORD?

(A) 3 (B) 4 (C) 6 (D) 7 (E) 8

---

**12.** 图中的风车星形是将一个直角三角形围绕它的一个顶点旋转而构成的。请问图中标记有点上的角之度数为何？

This pinwheel star is formed by rotating a right-angled triangle around one of its corners. What is the angle at each of the nine tips that are marked with dots?



(A) 30° (B) 40° (C) 45° (D) 50° (E) 60°

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**13.** 我有十二个容量都为 12 L 的油漆罐。已知其中一半的罐子恰装了一半的油漆、三分之一的罐子恰装了三分之一的油漆，剩下的罐子都装了六分之一的油漆。请问我总共有多少 L 的油漆？

I have twelve paint tins each capable of holding twelve litres. Half of them are half full. A third of them are a third full. The rest are one-sixth full. How many litres of paint do I have in total?

(A) 48 (B) 50 (C) 52 (D) 54 (E) 56

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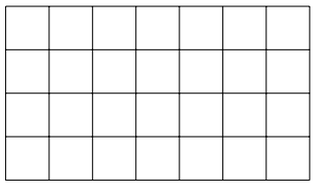
**14.** 请问有多少种方法可以将一个 $3 \times 1$ 的矩形放置在这个方格表内使得它可以完整地盖住三个小方格？

How many ways are there of placing a single $3 \times 1$ rectangle on this grid so that it completely covers three grid squares?



(A) 34 (B) 28 (C) 56 (D) 40 (E) 10

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**15.** 最接近星期二上午 10 点整之后 2017 分钟的时刻为

(A) 星期二下午 7：30 (B) 星期三上午 7：30 (C) 星期三下午 7：30 (D) 星期四上午 7：30 (E) 星期四下午 7：30

The time 2017 minutes after 10 am on Tuesday is closest to

(A) 7.30 pm Tuesday (B) 7.30 am Wednesday (C) 7.30 pm Wednesday (D) 7.30 am Thursday (E) 7.30 pm Thursday

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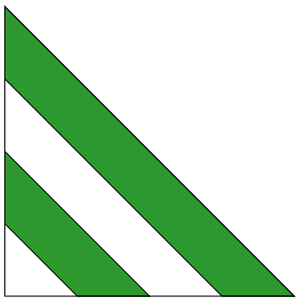
**16.** 三角形的两股都被斜线分割为四等分，如图所示。请问这个三角形的几分之几被涂上阴影？

The bottom and left side of this triangle are divided into 4 equal parts by the diagonal lines. What fraction of the large triangle is shaded?

(A) $\frac{5}{8}$ (B) $\frac{1}{2}$ (C) $\frac{3}{4}$ (D) $\frac{2}{3}$ (E) $\frac{3}{5}$



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**17.** 已知分数 $\frac{4}{n}$、$\frac{5}{n}$、$\frac{7}{n}$ 全都是最简分数。请问下列哪一个选项中的数可以是 $n$ 的值？

Each of the fractions $\frac{4}{n}$, $\frac{5}{n}$, $\frac{7}{n}$ is in its simplest form. Which of the following could be the value of $n$?

(A) 24 (B) 25 (C) 26 (D) 27 (E) 28

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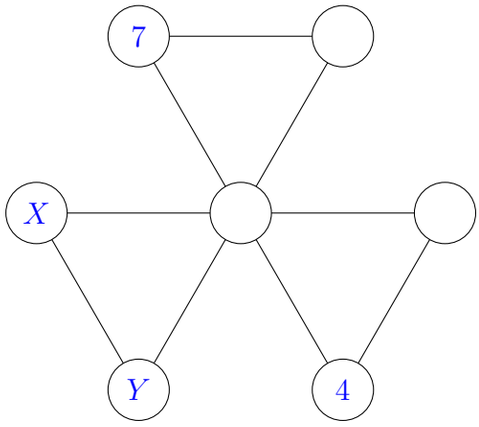
**18.** 将正整数 1、2、3、4、5、6、7 不重复地恰填入图中的七个圆圈内，使得所画出的三个三角形顶点上的数之和都相等。
现已填入其中两个数，如图所示。
请问 $X+Y$ 的值是多少？

The whole numbers from 1 to 7 are to be placed in the seven circles in the diagram. In each of the three triangles drawn, the sum of the three numbers is the same.
Two of the numbers are given.
What is $X+Y$?

(A) 5 (B) 6 (C) 7 (D) 8 (E) 9

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**19.** 小花、小桂与小宏玩疯狂拆砖电脑游戏，三人共有 2017 块砖要拆。他们按照以下方式分工：小花恰为小桂的二倍、小桂恰为小宏的二倍，且剩下的砖数量要尽可能地少。请问小花分配到要拆多少块砖？

Farhad, Greg and Huong were dismantling their marble madness machine and had 2017 marbles to share. They split them so that Farhad had exactly twice as many as Greg, and Greg had twice as many as Huong, with as few left over as possible. How many marbles were in Farhad's share?

(A) 1008 (B) 504 (C) 288 (D) 1344 (E) 1152

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**20.** 将大于 1 的整数按照如图所示的方式依序填入表格内。请问数 501 会出现在哪一竖列？

Whole numbers greater than 1 are arranged in a table in the pattern shown. In which column will the number 501 be found?

(A) A (B) B (C) C (D) D (E) E

| A | B | C | D | E |
|---|---|---|---|---|
| | | 2 | 3 | 4 |
| 5 | 6 | 7 | | |
| | | 8 | 9 | 10 |
| 11 | 12 | 13 | | |
| | | 14 | 15 | 16 |
| 17 | 18 | 19 | | |
| | ⋮ | | ⋮ | |

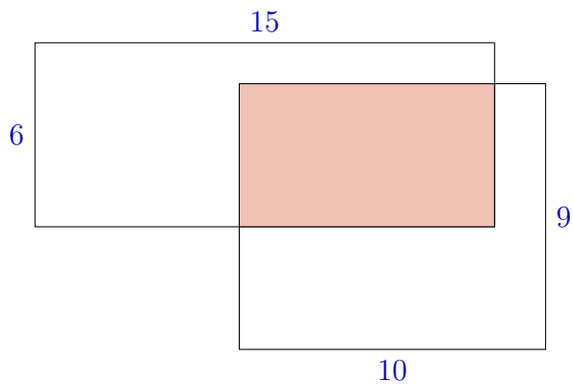
## 21-25 题，每题 5 分
## Questions 21 to 25, 5 marks each

**21.** 将两个矩形重迭造成三个面积相等的区域，原来的两个矩形分别是 6 cm × 15 cm 与 10 cm × 9 cm，如图所示。若阴影区域小矩形的每一边之长度都是整数 cm，请问阴影区域小矩形的周长为多少 cm？

Two rectangles overlap to create three regions, each of equal area. The original rectangles are 6 cm by 15 cm and 10 cm by 9 cm as shown. The sides of the smaller shaded rectangle are each a whole number of centimetres.

What is the perimeter of the smaller shaded rectangle, in centimetres?

(A) 24 (B) 28 (C) 30 (D) 32 (E) 36



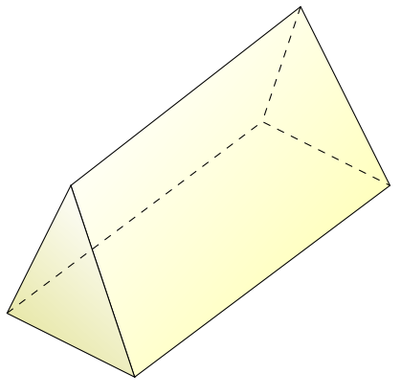

**22.** 一个回文数是由前往后读与由后往前读都是相同的数。已知数 131131 是一个回文数，且它的首二位数码 (13)、中间二位数码 (11)、末二位数码 (31) 都分别构成一个质数。请问共有多少个具有此性质的六位回文数？

A number is a palindrome if it reads the same forwards as backwards. The number 131131 is a palindrome; also the first pair of digits (13), the middle pair of digits (11) and the last pair of digits (31) are prime numbers. How many such 6-digit palindromes are there?

(A) 8 (B) 9 (C) 10 (D) 11 (E) 12

**23.** 将一个三角体柱用一个平面切一次而分成两片。请问这两片的面数之总和的最小可能值是多少？

A triangular prism is to be cut into two pieces with a single straight cut. What is the smallest possible total for the combined number of faces of the two pieces?



(A) 6 (B) 8 (C) 9 (D) 10 (E) 11

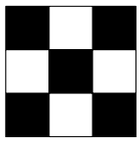
**24.** 小克与小思正在争论他们之间如何分配 120 mL 的汽水。
为了平息争论，他们的爸爸将小克杯中三分之一的汽水倒入小思的杯中，接着再将小思杯中三分之一的汽水倒入小克的杯中。此时他们两人杯中的汽水恰好一样多。
请问初始时小克的汽水比小思的汽水多或是少几 mL？

(A) 少 60 mL (B) 少 30 mL (C) 相等 (D) 多 30 mL (E) 多 60 mL

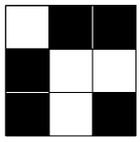
Ike and Seb were arguing over how 120 mL of soft drink had been shared between them.
To settle the argument, their dad poured one-third of Ike's drink into Seb's glass, and then he poured one-third of Seb's drink back into Ike's glass. Now they have an equal amount.
How much soft drink did Ike originally have compared to Seb?

(A) 60 mL less (B) 30 mL less (C) the same (D) 30 mL more (E) 60 mL more

**25.** 一个 $3 \times 3$ 方格表的小方格内都被涂上了黑色或是白色。
若其中任何一个 $2 \times 2$ 子方格表内每一种颜色都恰各占有二个小方格，则称这个方格表为均衡的，如第一个范例。
第二个范例则是不均衡的，因为右下角的 $2 \times 2$ 子方格表内有三个白色小方格。
若将旋转或翻转后不同的方格表视为不同，请问总共有多少个均衡的 $3 \times 3$ 方格表？



均衡的
balanced



不均衡的
unbalanced

A $3 \times 3$ grid has a pattern of black and white squares.
A pattern is called *balanced* if each $2 \times 2$ subgrid contains exactly two squares of each colour, as seen in the first example.
The pattern in the second example is *unbalanced* because the bottom-right $2 \times 2$ subgrid contains three white squares.
Counting rotations and reflections as different, how many balanced $3 \times 3$ patterns are there?

(A) 2 (B) 7 (C) 10 (D) 14 (E) 18

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**问题 26-30 的答案为 000-999 之间的整数，**
**请将答案填在答案卡上对应的位置。**

**第 26 题占 6 分，第 27 题占 7 分，第 28 题占 8 分，**
**第 29 题占 9 分，第 30 题占 10 分。**

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**For questions 26 to 30, shade the answer as an integer from 0 to 999 in the space provided on the answer sheet.**

**Question 26 is 6 marks, question 27 is 7 marks, question 28 is 8 marks, question 29 is 9 marks and question 30 is 10 marks.**

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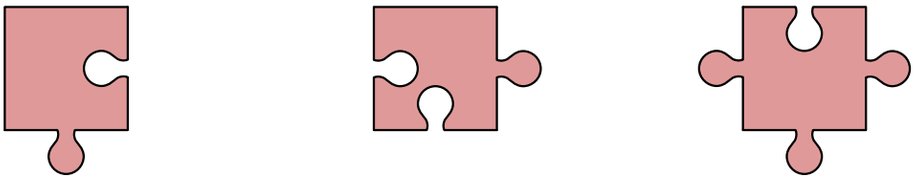
**26.** 从 0 到 9 的十个数码都各恰使用一次可构成两个五位数。请问这两个五位数之差的最小可能值是多少？

All of the digits from 0 to 9 are used to form two 5-digit numbers. What is the smallest possible difference between these two numbers?

---

**27.** 一块拼图的配件是由一张正方形硬纸板，在它的至少二条边上分别各有一个「插榫」或「接榫」的组合。
拼图配件分别有角落、棱边或内部的配件，如下图所示。



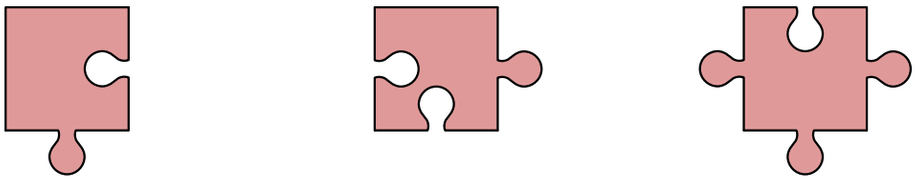
角落的配件
(两条直线边构成一直角)

棱边的配件
(有一条直线边)

内部的配件
(没有直线边)

若一块拼图的配件经过旋转 (不可以翻转) 后会与另外一块重合，则认定这两块配件的形状是相同的。请问总共有多少种不同可能形状的拼图配件？

A jigsaw piece is formed from a square with a combination of 'tabs' and 'slots' on at least two of its sides.
Pieces are either corner, edge or interior, as shown.



corner piece
(two straight sides at right angles)

edge piece
(one straight side)

interior piece
(no straight sides)

We treat two shapes as the same if one is a rotation of the other, without turning it over. How many different shapes are possible?

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**28.** 将数 129 逆序来读为 921，且这两个数之和为 1050，它可被 30 整除。请问总共有多少个三位数具有与它逆序的数之和可被 30 整除？

The reverse of the number 129 is 921, and these add to 1050, which is divisible by 30. How many three-digit numbers have the property that, when added to their reverse, the sum is divisible by 30?

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**29.** 我有非常多个玩具士兵。我可以将这些玩具士兵排成一个有许多行、许多列的矩阵。我注意到若我从中移除 100 个玩具士兵后，仍可以将剩下的玩具士兵恰排成一个减少 5 行、增多 5 列的矩阵。

请问我必须从初始的矩阵中移除多少个玩具士兵才能将剩下的玩具士兵恰排成一个减少 11 行、增多 11 列的矩阵？

I have a large number of toy soldiers, which I can arrange into a rectangular array consisting of a number of rows and a number of columns. I notice that if I remove 100 toy soldiers, then I can arrange the remaining ones into a rectangular array with 5 fewer rows and 5 more columns.

How many toy soldiers would I have to remove from the original configuration to be able to arrange the remaining ones into a rectangular array with 11 fewer rows and 11 more columns?

---

**30.** 小迈将至少二个连续正整数相乘而得到一个六位数 $N$。已知 $N$ 的首两位数码为 47 且 $N$ 的末两位数码为 74，请问小迈相乘时所使用的这些正整数之和是多少？

Mike multiplied at least two consecutive integers together. He obtained a six-digit number $N$. The first two digits of $N$ are 47 and the last two digits of $N$ are 74. What is the sum of the integers that Mike multiplied together?

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# Solutions – Junior Division

**1.** (Also MP1)
$2 + 0 + 1 + 7 = 10$,

hence (A).

**2.** The two angles on the left form a right angle: $20° + x° = 90°$.
Then $x° = 90° - 20° = 70°$,

hence (B).

**3.** The area is $3 \times 3.2 = 9.6\,\text{m}^2$,

hence (A).

**4.** The numbers counted are those that are 3 more than a multiple of 5. The units digit of this sequence alternates between 3 and 8. Of the answers given, only 48 has one of these units digits,

hence (B).

**5.** In each row of 4 circles, 3 are shaded, so that $\frac{3}{4}$ of the circles are shaded,

hence (D).

**6.** A number is divisible by 3 when the sum of its digits is a multiple of 3. The known digits add to $1 + 7 = 8$, which is 1 less than a multiple of 3. So the unknown digit will be 1 more than a multiple of 3, either 1, 4 or 7. Of these, only 1 is listed,

hence (A).

**7.** With 32 litres, it will run 4 times longer than with 8 litres. This is $4 \times 150 = 600\,\text{minutes} = 10\,\text{hours}$,

hence (D).

**8.** In kilograms, $0.15 + 0.78 + 1.35 + 0.68 = 2.96$,

hence (D).

**9.** (Also I4, S3)
100% of 1 is 1, and 1000% is 10 times greater, which is 10,

hence (C).

**10.** For the fraction to be between $7 = \frac{42}{6}$ and $8 = \frac{48}{6}$, the numerator must be between 42 and 48. For this, the number in the box must be between 39 and 45. Of the five numbers given, only 41 is in this range,

hence (E).

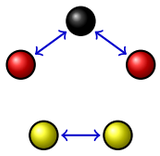
**11.** (Also I7, S6)
There are many ways to solve this with 4 swaps. For example,

$$\text{WORDS} \to \text{WORSD} \to \text{WOSRD} \to \text{WSORD} \to \text{SWORD}$$

This suggests that the answer is 4.

Suppose this isn't the smallest: some solution with three or fewer swaps changes WORDS to SWORD.

Each swap involves two letters, and each letter must be swapped at least once. When swapping five objects with these conditions, the only pattern of swaps possible is shown in the diagram: there are three swaps, one object is swapped twice and all other objects are swapped only once. In particular, there is a pair of objects that are swapped only with each other.



Such a pattern of swaps can't transform WORDS into SWORD, since there is no pair of letters which have only been swapped with each other. Consequently the smallest possible number of swaps is 4,

hence (B).

**12.** (Also S5)
The nine equal angles meeting at the centre of the pinwheel must each be $\frac{360}{9} = 40°$. Within each right-angled triangle, the angles add to 180°, so they are 40°, 90° and 50°, where 50° is the angle marked with a dot,

hence (D).

**13.** Six tins contain 6 L each. Four contain 4 L each. The two remaining contain 2 L each. Hence the volume is $6 \times 6 + 4 \times 4 + 2 \times 2 = 36 + 16 + 4 = 56$ litres,

hence (E).

**14.** (Also I8)
If the rectangle is horizontal, the leftmost square can be on any of the squares in the first 5 columns, making $5 \times 4 = 20$ ways.

If the rectangle is vertical, the topmost square can be in any of the first two rows, making $7 \times 2 = 14$ ways.

In total, there are $20 + 14 = 34$ ways,

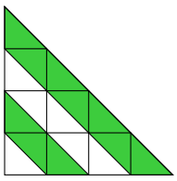
hence (A).

**15.** Divide $2017 \div 60 = 33\text{r}37$, so that 2017 minutes = 33 hours and 37 minutes. This is 1 day, 9 hours and 37 minutes. So 2017 minutes after 10 am on Tuesday is 7:37 pm on Wednesday,

hence (C).

**16.** *Alternative 1*

Drawing vertical and horizontal lines creates a grid of 16 congruent triangles, 10 of which are shaded. So the fraction is $\frac{10}{16} = \frac{5}{8}$,



hence (A).

*Alternative 2*

Choose a unit so that the base and left side each have length 4. Each shaded trapezium is the difference between two right-angled triangles, so their areas are $\frac{1}{2} \times 4 \times 4 - \frac{1}{2} \times 3 \times 3 = 3.5$ and $\frac{1}{2} \times 2 \times 2 - \frac{1}{2} \times 1 \times 1 = 1.5$. So the total shaded area is 5, out of a total area of 8,

hence (A).

**17.** The fraction $\frac{4}{n}$ is not in simplest form when $n$ is even. This rules out 24, 26 and 28.

The fraction $\frac{5}{n}$ is not in simplest form when $n$ is a multiple of 5. This rules out 25.

Checking 27, each of $\frac{4}{27}$, $\frac{5}{27}$ and $\frac{7}{27}$ are in simplest form, since 3 is the only prime factor of 27, but 3 is not a factor of 4, 5 or 7. Thus $n$ could only be 27,

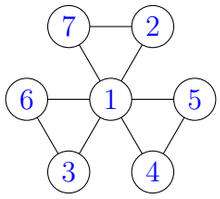
hence (D).

**18.** (Also UP22)

*Alternative 1*

Let $C$ be the number in the centre, so that the sum on each triangle is $T = X+Y+C$. Adding all three triangles gives $3T$, and this includes all 7 numbers, except that the centre number is counted three times. That is,

$$3T = 1 + 2 + 3 + 4 + 5 + 6 + 7 + 2C = 28 + 2C$$

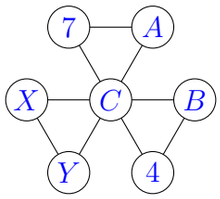



For $28 + 2C$ to be a multiple of 3, $C = 1$, 4 or 7. However 4 and 7 are already in the diagram, so that $C = 1$.

Then $3T = 28+2 = 30$, which means that $T = 10$ and $X+Y = 9$. The diagram on the right shows one of the two possible solutions with $X + Y = 9$,

hence (E).

*Alternative 2*




Label the diagram as shown and let $Q = X+Y$. Each triangle has the same total, so $Q = X + Y = 7 + A = B + 4$. Clearly $Q \geqslant 8$.

If $Q = 8$, then $B = 4$, which would repeat 4. So $Q \neq 8$.

If $Q = 9$, then $A = 2$ and $B = 5$. The remaining three numbers are 1, 3 and 6, so that $X$ and $Y$ are 3 and 6 (in either order) and $C = 1$. That is, there are two solutions when $Q = 9$.

If $Q = 10$, then $A = 3$ and $B = 6$. The remaining numbers are 1, 2 and 4. However, $X + Y = 10$, so that $Q = 10$ is not possible.

If $Q > 10$, then $B > 6$, which is not possible.

Consequently there are only two solutions, both with $X + Y = 3 + 6 = 9$,

hence (E).

**19.** For every 1 marble Huong has, Greg has 2 and Farhad has 4. So the total number of marbles shared is 7 times Huong's marbles. Since $2017 \div 7 = 288\text{r}1$, the most marbles Huong could have is 288, and then Farhad has $4 \times 288 = 1152$,

hence (E).

**20.** The pattern of columns repeats every 6 numbers. Also $501 \div 6 = 83\text{r}3$, which means that 501 will appear in the same column as 3, which is column D,

hence (D).

**21.** (Also UP26)
The $15 \times 6$ rectangle has area $90\,\text{cm}^2$, and the shaded area is half of this, which is $45\,\text{cm}^2$. This is also half of the $10 \times 9$ rectangle.
This shaded rectangle can then be $45 \times 1$, $15 \times 3$ or $9 \times 5$, but only the $9 \times 5$ rectangle will fit inside the $15 \times 6$ rectangle as shown. Consequently the perimeter of the shaded rectangle is $2 \times (5 + 9) = 28\,\text{cm}$,

hence (B).

**22.** The middle two digits must be 11, as they must be equal and form a prime number. The first two digits must be a prime number which also forms a prime number when the digits are reversed. There are nine such 2-digit primes:

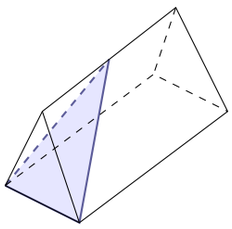
$$11, 13, 17, 31, 37, 71, 73, 79 \text{ and } 97$$

and so there are nine such 6-digit numbers:

$$111111, 131131, 171171, 311113, 371173, 711117, 731137, 791197 \text{ and } 971179,$$

hence (B).

**23.** (Also UP25)
The original prism has 5 faces. The cut will expose two totally new faces (either side of the cut) and any face it passes through will become two faces. So there are at least 7 faces in the final two polyhedra, and the minimum combined number of faces will occur when the cut passes through as few faces as possible.



The diagram shows that a cut along one edge and passing through two faces is possible, creating two pieces with a total of 9 faces.
If 7 faces were the minimum, they would be the two exposed by the cut and the 5 original faces. The plane would cut the surface of the polyhedron only through its edges. However there is no such plane: once a plane contains two or more edges of the prism, it is a face of the prism.
If 8 faces were the minimum, these would be the two faces exposed by the cut, 4 faces that are not cut and one face that is cut in two. Again, such a cut would contain two edges of the prism, so it is not possible.
Consequently the minimum number of faces in the two pieces is 9,

hence (C).

**24.** *Alternative 1*
The brothers both finish with 60 mL. We work backwards from there. Since one-third of Seb's drink was given away during the second pouring, this means that after the first pouring Seb had 90 mL and Ike had 30 mL. Since one-third of Ike's drink was given away during the first pouring, this means that Ike originally had 45 mL and Seb originally had 75 mL. Therefore Ike started with 30 mL less,

hence (B).

*Alternative 2*

Let $x$ and $120-x$ be the amounts of drink that Ike and Seb first received, respectively. After Ike gives one-third of his drink to Seb, Ike has $\frac{2x}{3}$ and Seb has the remaining $120-\frac{2x}{3}$. After Seb gives one-third of his drink back to Ike, Seb has $80-\frac{4x}{9}$ and Ike has the remaining $40+\frac{4x}{9}$. Setting these amounts equal gives

$$\begin{aligned} 40+\frac{4x}{9} &= 80-\frac{4x}{9} \\ \frac{8x}{9} &= 40 \\ x &= 45 \end{aligned}$$

Therefore Ike started with 45 mL and Seb started with 75 mL, so Ike started with 30 mL less,

hence (B).

**25.** (Also UP30)

*Alternative 1*

For each balanced pattern there is another which can be obtained by switching black and white throughout the entire grid. Hence we may assume the top-left square is black and double the number of such patterns. The top-left $2 \times 2$ subgrid can be coloured in three ways, with the grey squares yet to be determined:

(1) (2) (3)

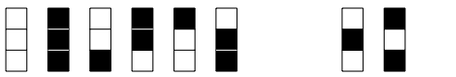
In case (1), $a$ and $b$ must both be black so the top-right $2 \times 2$ subgrid is balanced; the bottom row can alternate either black–white–black or white–black–white, hence there are 2 patterns for case (1). Since case (2) is the reflection of case (1) along a diagonal, there are also 2 patterns for case (2).

Case (3) allows two possibilities for $a$ and $b$: if $a$ is black and $b$ is white then the bottom row can again alternate in either of two ways, and if $a$ is white and $b$ is black then the bottom row must be black–white–white, hence there are 3 patterns for case (3). This gives 7 patterns in total when the top-left square is black, so doubling gives a total of 14 patterns,

hence (D).

*Alternative 2*

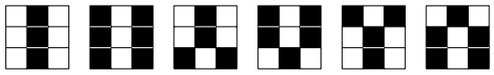
Consider the pattern as made up of 3 columns. There are 8 possibilities for the first column, which can either be non-alternating or alternating.



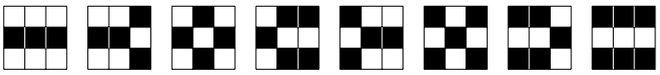
non-alternating alternating

In the six non-alternating possibilities for the first column, there are two (or more) adjacent squares in the column that are the same colour. To be balanced, the corresponding squares in the next column must be the reversed colour. Then every square

in the second column is forced to be the reverse of the corresponding square in the first column. For the same reason, the third column must be the reversal of the second column. Then each row alternates black–white–black or white–black–white. So there are 6 patterns where the first column is non-alternating:



In the two alternating possibilities for the first column, the next column must also be alternating, but there are two choices—it can either be the same or reversed. Likewise there are two choices for the third column. In all, there are 8 balanced patterns where the first column is alternating:



Combining these two cases, the number of balanced patterns is $6 + 8 = 14$,

hence (D).

**26.** (Also MP30, UP28, I19, S18)
The numbers must have different first digits, and so the smallest possible difference will be when the first digits differ by 1. Furthermore, the number formed by the remaining digits of the smaller number will be as large as possible and the number formed by the remaining digits of the larger number will be as small as possible.

The largest and smallest possible numbers formed from 4 digits are 9876 and 0123, so the two 5-digit numbers are 50123 and 49876 and their difference is $50123 - 49876 = 247$,

hence (247).

**27.** Denoting straight edges by 'E', slots by 'S' and tabs by 'T', each shape can be represented by a string of four of these symbols, subject to the following rules:

- There can be at most two E's.
- If there are two E's, they are adjacent.
- Cyclic permutations are considered equal: ESST = TESS = STES = SSTE.

The possible pieces are listed in the following table, where $\star$ represents 'T' or 'S':

| **Type** | **Form** | **Number** | **List** |
|---|---|---|---|
| Corner piece | E E$\star\star$ | $2^2 = 4$ | EESS EETS<br>EEST EETT |
| Edge piece | E$\star\star\star$ | $2^3 = 8$ | ESSS ETSS<br>ESST ETST<br>ESTS ETTS<br>ESTT ETTT |
| Interior piece | $\star\star\star\star$ | 6 | SSSS STST<br>SSST STTT<br>SSTT TTTT |
| **Total** | | **18** | |

There are 18 such pieces in total,

hence (18).

**28.** Let the number $N$ have digits $a$, $b$ and $c$, and let $N$ plus its reverse be $30k$, where $k$ is a whole number. Then

$$30k = 100a + 10b + c + 100c + 10b + a = 101(a + c) + 20b$$

This is divisible by 10 and $a \geqslant 1$, so $a + c = 10$. Then

$$\begin{aligned} 30k &= 1010 + 20b \\ k &= 33 + \frac{2}{3}(b + 1) \end{aligned}$$

Hence $b + 1$ is divisible by 3. That is, $b = 2$, 5 or 8.

For each of these 3 values of $b$, $a$ could be any digit from 1 to 9 and then $c = 10 - a$. In all there are 27 solutions: $129, 159, 189, 228, 258, 288, \ldots, 921, 951$ and $981$,

hence (27).

**29.** (Also I29)

*Alternative 1*

Suppose that the original number of rows is $r$ and the original number of columns is $c$, so that I have $r \times c$ toy soldiers altogether. Then the condition of the problem can be written as

$$\begin{aligned} & r \times c - 100 = (r - 5) \times (c + 5) \\ \Rightarrow \quad & rc - 100 = rc + 5r - 5c - 25 \\ \Rightarrow \quad & 5c - 5r = 75 \\ \Rightarrow \quad & c - r = 15 \end{aligned}$$

The number of toy soldiers I would have to remove from the original configuration to be able to arrange the remaining ones into a rectangular array with 11 fewer rows and 11 more columns is

$$\begin{aligned} r \times c - (r - 11) \times (c + 11) &= rc - (rc + 11r - 11c - 121) \\ &= 11c - 11r + 121 = 11(c - r) + 121 \end{aligned}$$

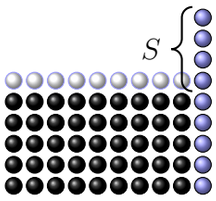
However, since we have already deduced that $c - r = 15$, the answer is $11 \times 15 + 121 = 286$,

hence (286).

*Alternative 2*

Consider the consequences of removing one row of soldiers and adding one column.

This creates a surplus if the new height is less than the old width. The surplus is the difference between the initial width and the final height. Let this surplus be $S$.




When the next row is made into a column, the surplus is two more than before, since the row is 1 soldier wider and the column is 1 soldier shorter. So the surplus is $S + 2$.

Proceeding in the same way, surpluses from the next moves are $S + 4$, $S + 6$ and $S + 8$.

After 5 rows have been moved, the total surplus is 100, and so the average surplus is 20. That is, the surpluses are $16, 18, 20, 22$ and $24$.

If we continue to 11 rows, the total surplus is

$$16 + 18 + 20 + 22 + 24 + 26 + 28 + 30 + 32 + 34 + 36 = 286$$

hence (286).

**30.** (Also S27)

Since 74 is not divisible by 4 and $N - 74$ is divisible by 4, the number $N$ is not divisible by 4. Consequently, exactly one of the consecutive numbers is even, so there are either two or three consecutive numbers in the product.

For $N = a(a+1)$, consider the units digit of $N$. The units digits of $0 \times 1$, $1 \times 2$, ..., $9 \times 10$ are 0, 2, 6, 2, 0, 0, 2, 6, 2, and 0, respectively. None of these are 4, so there is no solution with $N = a(a+1)$.

Consider $N = a(a+1)(a+2)$. Since $N$ is not a multiple of 4, $a$ is odd. When the units digit of $a$ is $1, 3, 5, 7$ or 9, the units digit of $N$ is $6, 0, 0, 4$ or 0, respectively so the units digit of $a$ is 7.

Now, $N \approx a^3$ and we can note that $70^3 = 343\,000 < N < 512\,000 = 80^3$. This suggests that $a = 77$. More precisely, if $a \leqslant 67$, then $N \leqslant 67 \times 68 \times 69 < 70^3$, whereas if $a \geqslant 87$, then $N \geqslant 87 \times 88 \times 89 > 80^3$. Hence the only possibility is $a = 77$.

Checking, $77 \times 78 \times 79 = 474\,474$, so the numbers that Mike multiplied together are 77, 78 and 79. Then $77 + 78 + 79 = 234$,

hence (234).

# Answers

| Question | Middle Primary | Upper Primary | Junior | Intermediate | Senior |
|---|---|---|---|---|---|
| 1 | A | C | A | C | A |
| 2 | C | B | B | A | A |
| 3 | E | E | A | C | C |
| 4 | A | A | B | C | D |
| 5 | D | D | D | C | D |
| 6 | E | A | A | E | B |
| 7 | A | B | D | B | C |
| 8 | C | B | D | A | E |
| 9 | D | D | C | A | C |
| 10 | D | A | E | D | D |
| 11 | A | E | B | E | C |
| 12 | B | B | D | C | E |
| 13 | B | E | E | E | E |
| 14 | C | D | A | D | B |
| 15 | E | B | C | D | C |
| 16 | C | A | A | B | B |
| 17 | B | D | D | B | D |
| 18 | A | D | E | E | D |
| 19 | E | E | E | D | A |
| 20 | C | B | D | E | B |
| 21 | E | D | B | C | E |
| 22 | C | E | B | A | B |
| 23 | D | B | C | D | C |
| 24 | C | C | B | E | D |
| 25 | D | C | D | E | E |
| 26 | 337 | 28 | 247 | 807 | 13 |
| 27 | 13 | 60 | 18 | 648 | 234 |
| 28 | 4 | 247 | 27 | 56 | 56 |
| 29 | 468 | 18 | 286 | 286 | 360 |
| 30 | 247 | 14 | 234 | 99 | 510 |